

Transduced Valence: From Neural Effective Drive to ATP-Powered Action

A cross-scale Self Aware Networks model of metabolic support, motor work, and sensory return

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Abstract

A spoken syllable or a reach begins in millisecond neural events but becomes real only through a longer chain. Membrane currents and calcium alter synaptic work; glycolysis and mitochondria supply ATP; blood flow later delivers oxygen and substrate; motor neurons recruit muscle; ATP hydrolysis powers contraction; and the resulting sound or movement returns as new sensory evidence. These mechanisms are usually measured separately. The open problem is how to test whether one distributed neural relation is transformed across them without pretending that voltage, ATP flux, blood flow, force, and felt value are the same variable.

This paper develops **transduced valence** as a typed causal coordinate for that problem. A declared receiver integrates source amplitude, phase, sign, delay, geometry, inhibition, and its own state. The action-relevant component of that relation is called SAN electrical valence only after the operation is specified. ATP then sustains ionic restoration, vesicle cycling, cellular recovery, and muscular work. The recruited route gives that available biochemical power an action-relative sign; the resulting force changes the organism or world. Each interface has its own units, delay, state, and uncertainty. The proposed invariant is therefore not a conserved scalar but a testable causal direction: receiver-relative neural drive should improve prediction of energetic demand, delayed vascular recruitment, motor work, and returned evidence beyond disconnected descriptions of those layers.

A deterministic matched-input construction demonstrates identifiability under declared assumptions. Across 6,000 constructed trials and held-out blocks, adding receiver-relative neural valence raises motor-recruitment prediction from an R-squared of 0.078 to 0.999; typed models similarly improve force and sensory-return prediction. A null world correctly shows no useful gain from an added field feature, while a constructed field-positive world recovers the injected incremental term. These are model checks, not biological effect estimates.

The paper also isolates a harder optional hypothesis: endogenous neural electric or magnetic fields may contribute directly to mitochondrial ATP production after calcium, ions, substrate, oxygen, temperature, and stimulation energy are controlled. Existing ephaptic and applied-field studies do not establish that mechanism because their preparations, amplitudes, frequencies, geometries, and durations differ. The field term is therefore separable, low-prior, and removable without weakening the established neural-to-metabolic-to-motor chain.

Keywords: Self Aware Networks; Phase Wave Differential; electrical valence; mitochondria; ATP; neurovascular coupling; ephaptic coupling; motor units; muscle energetics; action-perception loop

1. The missing bridge is between a neural pattern and physical work

Suppose I decide to speak. Before the sound exists, many neural populations are active. Some encode the situation, some prepare candidate words, some regulate breathing and posture, and some influence laryngeal and articulatory motor pools. But activity in the brain is not yet acoustic pressure in the air. Therefore a complete account must explain how one distributed neural pattern becomes effective at selected receivers, how those receivers recruit cellular energy, how descending signals become muscular force, and how the resulting sound returns as new evidence.

The same problem appears in vision and reaching. A visual relation becomes meaningful when it changes what the organism can select, predict, or do next. But movement is metabolically and mechanically expensive. Therefore a theory that ends with a decoded brain pattern stops before the pattern has acquired its full biological consequence.

Neuroscience already contains the pieces of this chain. Action potentials propagate through excitable cellular compartments. Calcium helps couple electrical activity to transmitter release and mitochondrial metabolism. ATP supports ion pumps and synaptic-vesicle cycling. Neural and glial signals recruit local blood flow. Motor-unit discharge predicts force and energetic cost. What is missing is a common experimental grammar for asking whether a particular neural differential is transformed across those interfaces while retaining an action-relative direction and effective magnitude.

This paper develops that grammar. I call the joined operation **transduced valence** only after its problem is clear: a distributed difference must become effective at a receiver, acquire metabolic support, produce or prevent work, and return a consequence that changes the next network state.

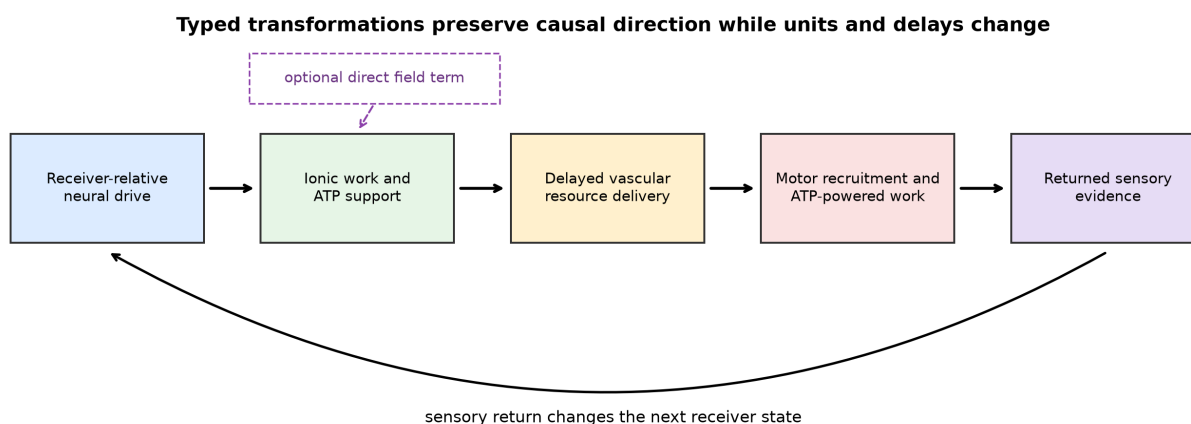


Figure 1. The cross-scale architecture. Solid arrows show the main established and testable chain. The dashed field term is an optional incremental hypothesis, not a premise of the paper.

2. Four measurements that should not be collapsed

The word *activity* can conceal several different measurements. An electrode can measure membrane voltage, spikes, or a field potential. An ATP sensor can measure concentration or production near a terminal. Functional imaging can measure blood oxygenation or flow. Electromyography and force transducers can measure motor-unit activity and mechanical output. These variables belong to one causal sequence, but no one measurement substitutes for the others.

One measurement boundary is especially useful. Diffusion tensor imaging estimates directional water diffusion and helps infer tissue microstructure and probable fiber orientation [1]. It does not measure the fast-to-slow energetic response considered here. BOLD fMRI, fNIRS, perfusion, vessel imaging, oxygen measurements, and metabolic sensors instead observe different projections of neurovascular and metabolic change. Structural route, neural event, ATP flux, hemodynamic response, and force must remain separately identified.

The four principal layers in this paper are therefore:

Layer	Example measurement	Characteristic role
Neural electrical	membrane voltage, spikes, phase, LFP, receiver drive	selection and effective signaling
Cellular metabolic	ATP production, ATP concentration, oxygen consumption, mitochondrial potential	energetic support and state regulation
Neurovascular	vessel diameter, cerebral blood flow, oxygenated/deoxygenated hemoglobin	delayed delivery of substrate and oxygen
Mechanical	motor-unit discharge, calcium-dependent activation, ATP turnover, force, sound	physical realization and environmental change

The paper seeks lawful transfer functions among these layers rather than one universal scalar that pretends they are already the same thing.

3. A firing neuron undergoes a multiscale state transition

An action potential is often represented as a binary output, but the living event recruits more than an abstract one. In neocortical pyramidal cells, axonally initiated action potentials can propagate back into somatic and dendritic compartments [2]. Their penetration and calcium consequences depend on cell type, compartment, voltage, and activity pattern. At terminals, calcium and the synaptic-vesicle cycle connect electrical activity to chemical release and energetic cost.

De Polavieja and colleagues provide a direct bridge from recent input to spike shape. Under naturalistic conductance stimulation in rat cortical pyramidal neurons, somatic action-potential waveforms reliably carried information about roughly the preceding 50 milliseconds of conductance history, with more information than instantaneous spike rate in that preparation [20]. The finding does not make waveform a universal code, but it establishes that the all-or-none description is incomplete when the question concerns the living event's history and downstream effect.

Rangaraju, Calloway, and Ryan measured presynaptic ATP and showed that electrical activity imposes substantial local demand, with the synaptic-vesicle cycle providing a major cost. Glycolysis and mitochondrial function were recruited to match that demand [3]. Ashrafi and colleagues then showed that axonal mitochondria use the brain-enriched MICU3 regulator to take up calcium during activity and accelerate ATP production under oxidative conditions [4]. Yang and colleagues identified a slower activity-linked route in which Rheb enters mitochondria and promotes pyruvate-dehydrogenase activity and ATP production [5].

Therefore firing changes the conditions for later firing. On millisecond timescales, voltage, conductance, calcium, and release change. Over seconds and minutes, pumps, metabolic pathways, mitochondrial recruitment, and substrate availability matter. With suitable bursts and repeated activity, signaling can reach transcription and protein synthesis. A neuron is consequently a stateful biological transformation system whose output reflects morphology, recent input, ionic state, release history, metabolic capacity, and current receiver relations.

That point supplies the cellular foundation for the Action Potential Magnitude Effect and PWD. The question is not whether every action potential has a freely varying peak amplitude. It is how duration, waveform, burst structure, phase, terminal state, release, and downstream consequence vary within the living cell's changing state.

4. The SAN genealogy reaches this question in stages

The present model developed by repeatedly returning to one problem: how a neural difference becomes an embodied consequence. A note preserving a dialogue from before September 2011 describes a loop in which senses form brainwaves, brainwaves change muscles and sound, and the changed action is sensed again [21]. Its authored date and its later 2022 public Git fixation are different provenance facts; it establishes an early source ancestor, not a 2011 version of the equations below.

A public Neo Mind Cycle dialogue uploaded in August 2012 describes EEG-driven light and sound as a two-way conversation. Feeling-associated coordinated neural changes alter the returned media, which the participant then sees and hears [22]. That source contributes the measurable-state and sensory- return parts of the chain. It does not contain the later PWD or transduced-valence vocabulary.

The April 2017 Neural Lace Podcast makes the information-energy bridge deliberate. It names a proposed **Alternating Transmission Protocol (ATP)** and immediately pairs the acronym with adenosine triphosphate as the cell's energy currency. The same episode treats cells as coincidence detectors and proposes transmitting the difference between a present pattern and a desired pattern [23]. The analogy does not show that neurons literally implement Internet protocols or that ATP concentration selects a communication mode. It does establish that communication, coincidence, difference, and biological energy were already being developed as one research problem.

An authored July 2017 source then compares fish electric sensing and bat echolocation, treats neurons as oscillatory send-and-return systems, and contrasts tonic context with phasic change. Its current public Git form was fixed in 2022 [15]. By June through August 2022, the APME notes connect potassium-channel regulation, action-potential duration, calcium-channel opening, and variable release to the signal presented to later cells [16]. *Phase Shifts as Electromagnetic Wave Valence* then explicitly joins phase change, relative electrical valence, neural-array recruitment, affect, and behavior [17].

The bat studies supply a concrete preparation. SAN notes incorporate pre-vocal frontal and auditory oscillatory changes and interpret them as a PWD-to-action-to-return bridge [18,19]. Those historical sources do not contain the equations below. They establish the developing question: can a building receiver-relative pattern recruit cellular support and bodily work, then return a consequence that changes the next network state? The current paper supplies the typed formalization and the tests.

5. Neural electrical valence is effective drive at a receiver

Imagine several inputs reaching a receiving population. Their effects depend on sign, amplitude, phase, delay, geometry, synaptic gain, inhibition, and the receiver's current state. Inputs arriving in a compatible relation can add effectively; opposing inputs can cancel, redirect, or prevent threshold crossing. Therefore raw power or spike count alone cannot specify which candidate becomes effective.

For a declared receiver (r), a compact complex-valued representation of the transformed source signals is

$$Z_N^{(r)}(t) = \sum_{i=1}^N H_{ir}(t) A_i(t) e^{i\phi_i(t)}, \quad V_N^{(r)}(t) = \text{Re} \left[e^{-i\theta_r(t)} Z_N^{(r)}(t) \right]. \quad (1)$$

Here $\mathbf{H}_{ir}$ transforms source i into the receiver's frame, including sign, delay, gain, geometry, and any required unit conversion. $\mathbf{A}_i$ and ϕ_i describe the declared source amplitude and phase, and θ_r is the receiver reference. The phasor is an analytic representation unless an experiment identifies it with a specific physical field. $V_N^{(r)}$ is **SAN electrical valence**: the signed component of distributed activity that is effective along the receiver's current action-relevant direction. The full companion estimator separates amplitude budget, resultant magnitude, alignment, and cross-fitted context; Equation (1) is this paper's explanatory form.

An electrical component of a Phase-Wave Differential is the unexpected change in this drive relative to the state predicted for the same context:

$$D_N^{(r)}(t) = V_N^{(r)}(t) - \hat{\mathbb{E}}_{-g} \left[V_N^{(r)}(t) \mid \mathbf{c}_t, \mathbf{h}_t \right]. \quad (2)$$

The expectation is learned outside the held-out animal, session, or group (g). The PWD therefore names a consequential departure from a tonic reference, not a synonym for oscillation, coherence, or high-frequency power.

6. Neural activity supplies electrical drive; ATP sustains and realizes the route

The electrical valence defined in Equations (1) and (2) arises from neural currents reaching a declared receiver with particular signs, timings, amplitudes, transfer paths, and receiver state. Coincident compatible inputs can build effective drive; opposing relations can reduce or redirect it. ATP is not inserted as the source of that instantaneous electrical relation. It is required to maintain ion gradients, recycle vesicles, support cellular recovery, and power the contraction that eventually realizes a selected motor route.

ATP concentration, ATP production, ATP turnover, and biochemical free-energy flux are distinct. The quantity most closely related to available biochemical power is

$$P_{\text{ATP}}^{(c)}(t) = J_{\text{ATP}}^{(c)}(t) \left[-\Delta G_{\text{ATP}}^{(c)}(t) \right], \quad (3)$$

where $J_{\text{ATP}}^{(c)}$ is ATP hydrolysis flux in compartment c , and $-\Delta G_{\text{ATP}}$ is the free energy made available per mole under the measured cellular conditions. This biochemical power is nonnegative when ATP hydrolysis supplies work. Action direction is supplied separately by the recruited route.

The selected biological route supplies that direction. Define action-relative metabolic valence as

$$V_M^{(r,c)}(t) = s_r(t) g_{rc}(t) P_{\text{ATP}}^{(c)}(t), \quad s_r \in \{-1, +1\}, \quad g_{rc} \geq 0. \quad (4)$$

The sign s_r belongs to the declared action coordinate and recruited route: agonist versus antagonist, contraction versus relaxation, approach versus withdrawal, or effective excitation versus inhibition. The route-specific gain g_{rc} carries whatever conversion is required for the declared output; it is dimensionless only when V_M is reported as signed biochemical power. **SAN metabolic valence** is ATP-powered capacity oriented by the neural and mechanical route that uses it. Its sign belongs to the action coordinate rather than to ATP itself.

The general cross-scale claim is a sequence of state-dependent transformations:

$$V_{k+1}(t + \tau_k) = \mathcal{T}_k(V_k(t), \mathbf{z}_k(t)) + \epsilon_k(t), \quad (5)$$

where each interface k has its own units, delay τ_k , state variables $\mathbf{z}_k$, and uncertainty. What persists is not numerical identity. It is an action-relative direction and an effective magnitude whose transformations can be estimated, perturbed, and followed.

7. A separable harder test: can endogenous neural fields add anything?

The main chain does not require a direct field-to-mitochondria effect. The established baseline couples activity to ATP through calcium, ionic work, energy sensing, substrates, and mitochondrial membrane energetics. A deliberately broad baseline is

$$J_{\text{ATP}}^N(t) = f([\text{Ca}^{2+}]_m, [\text{ADP}], [\text{AMP}], O_2, \mathbf{S}, \Delta\Psi_m, W_{\text{pump}}, W_{\text{ves}}), \quad (6)$$

where $\mathbf{S}$ describes metabolic substrates, $\Delta\Psi_m$ is mitochondrial membrane potential, and the work terms include restoration of ion gradients and vesicle cycling. This is the baseline pathway through which action potentials and synaptic work change energetic demand and mitochondrial response.

Neural tissue also generates local electric and magnetic fields. In cortical slices, physiological extracellular fields produced somatic membrane-potential changes smaller than 0.5 millivolts and could entrain spikes, particularly for slow fluctuations below 8 Hz [6]. That establishes a neural ephaptic effect; it does not establish a mitochondrial effect.

Ito and colleagues provide a different comparator. They applied 10-microtesla magnetic pulses of 4 milliseconds in a repeating 1-8 Hz sequence for prolonged exposures to NARP and control cybrids [14]. ATP-related effects were delayed and strongest in a mitochondrial-disease cybrid preparation; control-cybrid effects were absent or modest. The applied amplitude, frequency, duration, geometry, cell type, and endpoint do not match an endogenous action-potential or PWD field. The study makes a field-sensitive mitochondrial experiment conceivable, but it cannot be used as evidence that the endogenous neural mechanism already exists.

The incremental question can nevertheless be made estimable. Let electric and magnetic features be calibrated or standardized separately within a declared preparation:

$$J_{\text{ATP}}^N(t) = f(\text{established ionic and metabolic variables}) + \kappa_E \tilde{\Phi}_E^{(m)}(t) + \kappa_B \tilde{\Phi}_B^{(m)}(t) + \epsilon(t), \quad (7)$$

with

$$\tilde{\Phi}_X^{(m)}(t) = \frac{\int_0^W w_X(\tau) \mathbf{X}_{\text{local}}(t - \tau) \cdot \mathbf{n}_{X,m} d\tau - \mu_X}{\sigma_X}, \quad X \in \{E, B\}. \quad (8)$$

Here $\mathbf{X}$ denotes the measured electric or magnetic field, $\mathbf{n}_{X,m}$ a declared orientation, w_X a prespecified temporal kernel, and μ_X, σ_X training-set location and scale. Standardization makes the feature dimensionless; κ_E and κ_B retain the units of ATP flux. The coefficients are estimated only after calcium, ADP/AMP, oxygen, substrate, temperature, membrane voltage, extracellular ions, glial state, and stimulation energy are modeled.

The analysis should begin with a skeptical prior centered near zero, a frozen minimum useful effect, and separate electric and magnetic perturbations. Reproducibly nonzero coefficients are insufficient by themselves: a selective field perturbation must remove the predicted ATP component, and a geometry- and amplitude-matched restoration must rescue it without changing the matched ionic and metabolic covariates. A negligible coefficient leaves the established electrochemical-metabolic route, receiver-relative neural-valence model, and action pathway unchanged.

8. Blood flow is a slower energetic response to local activity

Neural tissue cannot maintain signaling without oxygen and substrate. Local activity changes demand, but blood cells and vessels move more slowly than action potentials. Therefore the fast neural event and the slower hemodynamic response should be represented with an explicit delay rather than treated as simultaneous copies.

Let $Q_r(t)$ describe local metabolic demand estimated from synaptic, spiking, ionic, and ATP variables in region r . A hemodynamic measurement $H_r(t)$ can be written as

$$H_r(t) = \int_0^\infty K_r(\tau) Q_r(t - \tau) d\tau + \eta_r(t), \quad (9)$$

where K_r is a region-, vessel-, state-, and modality-specific response kernel. BOLD fMRI, fNIRS, blood-flow imaging, and oxygen measurements observe different projections of H_r .

The biology is distributed. Neurons, interneurons, astrocytes, endothelial cells, smooth muscle, and pericytes can participate through preparation-specific pathways [7,8]. Isaacs and colleagues provide an especially relevant bridge: thin-strand pericytes sensed local energetic conditions and relayed KATP-dependent electrical signals toward upstream arterioles, helping recruit blood flow [9]. This supplies a complementary vascular example: biological tissue translates local energy need into an electrical signal that travels through another cellular network to control resource delivery.

In the transduced-valence account, neurovascular coupling supplies a delayed support loop. A neural differential changes local energetic demand; metabolic and vascular cells register that demand; blood flow changes the available oxygen and substrate; and those resources alter which neural states can be sustained next.

9. Bats expose the transition from preparation to vocal action

Vocalization is a useful case because the neural preparation, muscle action, sound, and returned auditory consequence can all be measured. Weineck and colleagues recorded frontal and striatal activity in three bats and found pre-vocal beta- and high-gamma-related patterns that predicted whether the upcoming call served echolocation or communication [10]. Garcia-Rosales and colleagues recorded frontal and auditory cortex in three bats and found predominant frontal-to-auditory directionality during spontaneous and pre-vocal periods, followed by call- and frequency-dependent changes after vocalization, including auditory-to-frontal return associated with echolocation [11]. The latter authors explicitly state that the relationship between the measured oscillations and vocal production remains correlational; their data do not by themselves prove that an LFP initiates or plans the call.

These results provide the temporal architecture needed by the model: preparation precedes action, action changes the world, and the changed sound returns through sensory pathways. They also prevent one frequency-band slogan from replacing the mechanism. Beta and gamma relationships vary by region, depth, call type, epoch, and analysis. The relevant variable is which distributed relation becomes effective at the receiver that prepares or realizes the call.

SAN electrical valence names that effective receiver drive. But the field potential does not push a vocal fold directly. Therefore the physical route must continue through descending pathways, brainstem and spinal or cranial motor neurons, neuromuscular transmission, muscle-fiber calcium, ATP-powered cross-bridge cycling, respiratory and laryngeal mechanics, acoustic output, and auditory return.

10. Neural drive, ATP turnover, and force form a measurable chain

The complete action route can be stated before another term is introduced:

$$V_N(t) \longrightarrow R_{MU}(t) \longrightarrow C_{Ca}(t) \longrightarrow J_{ATP}(t) \Delta G_{ATP}(t) \longrightarrow F(t) \longrightarrow \mathbf{y}_{\text{return}}(t + \tau). \quad (10)$$

R_{MU} represents motor-unit recruitment and discharge, C_{Ca} muscle activation, F action-relative force, and $\mathbf{y}_{\text{return}}$ the resulting acoustic, proprioceptive, tactile, vestibular, visual, and internal sensory change.

The middle of this chain is already measurable. Christie, Foulis, and Kent combined indwelling EMG and phosphorus magnetic-resonance spectroscopy across voluntary contraction intensities. Motor-unit discharge rate and ATP cost both increased with intensity, and discharge rate explained a large fraction of ATP-cost variability in their sample [12]. Del Vecchio and colleagues found that motor neuron recruitment speed and maximal discharge were strongly related to rapid force development in human contractions [13].

These findings support a transduction relationship rather than numerical identity. A muscle's force also depends on fiber type, recruitment order, calcium handling, length, velocity, fatigue, temperature, mechanics, and history. A useful action model is therefore

$$F_a(t) = \mathcal{M}_a(R_{MU,\leq t}, C_{Ca,\leq t}, P_{ATP,\leq t}, \mathbf{z}_{\text{muscle},t}), \quad (11)$$

where action axis a determines sign and $\mathbf{z}_{\text{muscle}}$ contains the mechanical and physiological state. Transduced valence is the joined estimate of how a selected neural route changes this action-relative capacity and its returned consequence.

11. Experimental program

11.1 Isolate an incremental mitochondrial field term

The decisive cellular experiment compares population patterns with the same per-cell spike count, mean rate, calcium load, substrate, oxygen, temperature, and stimulation energy but different phase alignment and extracellular field geometry. Neurons would express compartment-specific voltage, calcium, mitochondrial-potential, ATP-production, and ATP-concentration sensors. Extracellular arrays would measure the local field, and oxygen-consumption measurements would supply an independent metabolic endpoint.

At least four conditions are needed:

1. phase-dispersed firing with a fixed per-cell spike sequence;
2. phase-aligned firing using the same per-cell sequence;
3. a calibrated applied field reproducing the measured field while synaptic and spiking routes are controlled; and
4. a field-reducing or geometry-altering intervention that preserves the matched cellular firing pattern as closely as possible.

Extracellular potassium, pH, glial activity, ADP/AMP, oxygen, and calcium must be measured because coherent firing can change all of them. Electric and magnetic terms should be estimated separately. Applied-field amplitudes must match the endogenous preparation rather than borrowing a therapeutic field regime.

Let $\mathbf{M}_0$ contain the established variables in Equation (6), and let $\mathbf{M}_1$ add the prespecified field features in Equation (8). For held-out cells, cultures, or animals,

$$\Delta L_{EM} = L(\mathbf{M}_0) - L(\mathbf{M}_1), \quad (12)$$

must exceed a frozen useful-effect threshold. A selective field perturbation should then remove the predicted ATP component, and a geometry-matched field restoration should rescue it without changing the matched spike and calcium variables. The experiment must also include a field-null analysis: an added field feature should not improve prediction in a preparation where the incremental coupling is absent. The loss, grouping level, split, minimum useful effect, and electric-versus-magnetic tests are frozen before the final held-out evaluation.

11.2 Follow neural drive into human muscle energetics

A second experiment combines high-density EMG, motor-unit decomposition, force, ultrasound or mechanomyography, and phosphorus MRS during graded and rapid contractions. Candidate neural-valence variables are estimated upstream where feasible or from motor-pool inputs. The analysis asks whether receiver-relative timing and recruitment improve prediction of ATP turnover and force beyond motor unit count, mean discharge rate, force history, fatigue, and muscle state.

Agonist and antagonist muscles are measured together so the sign in Equation (4) comes from the actual action coordinate. The strongest result is a selective timing or route perturbation that changes predicted ATP cost and force, followed by restoration of the relevant relation.

11.3 Connect fast neural events to slower vascular support

A third experiment combines electrophysiology or MEG/EEG with fNIRS, BOLD, perfusion, or direct vascular imaging. The model uses Equation (9) to preserve the hemodynamic delay. It tests whether the energetic consequences of a declared neural differential predict local vascular recruitment beyond conventional power, firing, task, motion, carbon-dioxide, and systemic physiology variables.

The three experiments form one ladder. The cellular study isolates any incremental field term. The muscle study tests neural-to-metabolic-to-mechanical transduction. The vascular study tests delayed resource delivery. No single experiment is required to stand in for the other two.

12. Deterministic identifiability demonstration

Before biological data exist, a constructed system can test whether the proposed variables are mathematically distinguishable under known assumptions. The accompanying script fixes the random seed and creates 3,000 paired source patterns, or 6,000 total trials. Each pair has exactly the same per-source amplitude/rate budget and the same calcium, ADP, oxygen, substrate, temperature, pump, vesicle, and muscle-state covariates. Only phase organization changes between the aligned and dispersed member. Fourteen of twenty blocks fit the models; six blocks remain held out.

The construction contains two ATP worlds. The **field-null world** has no incremental field term. Adding the field feature changes held-out R^2 from 0.55648 to 0.55629, correctly providing no useful gain. The **field-positive world** contains a small known term; adding the feature raises held-out R^2 from 0.418 to 0.647 and recovers the injected dependency. This is a recovery check, not evidence that the dependency exists in biology.

The other interfaces behave as designed. A model using only rate budget and context explains 0.078 of held-out motor-recruitment variance; adding receiver-relative neural valence raises R^2 to 0.999. A disconnected force model scores 0.082, while the typed neural/ATP/muscle model scores 0.976. A disconnected sensory-return model scores 0.442, while a model containing force, context, and neural valence scores 0.982. Removing the neural-to-motor interface collapses the aligned-versus- dispersed return gap from 0.371 to 0.000; matched restoration returns it to 0.371.

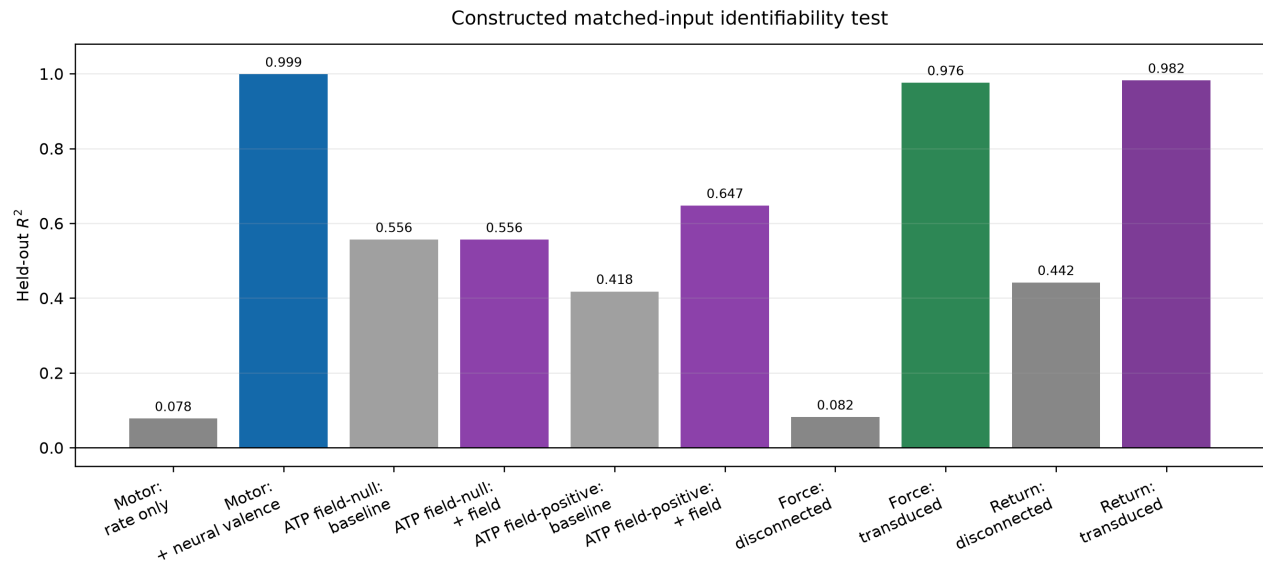


Figure 2. Held-out results from the constructed matched-input test. Near-perfect values reflect a deliberately recoverable synthetic system, not biological performance. The field-null comparison is the key negative control.

The illustration also keeps timescales explicit. A phasic neural departure can change local demand and motor recruitment quickly, while vascular support follows through a slower response kernel and the action's sensory result returns after an interface-specific delay.

One fast-to-slow-to-return sequence with explicit delays

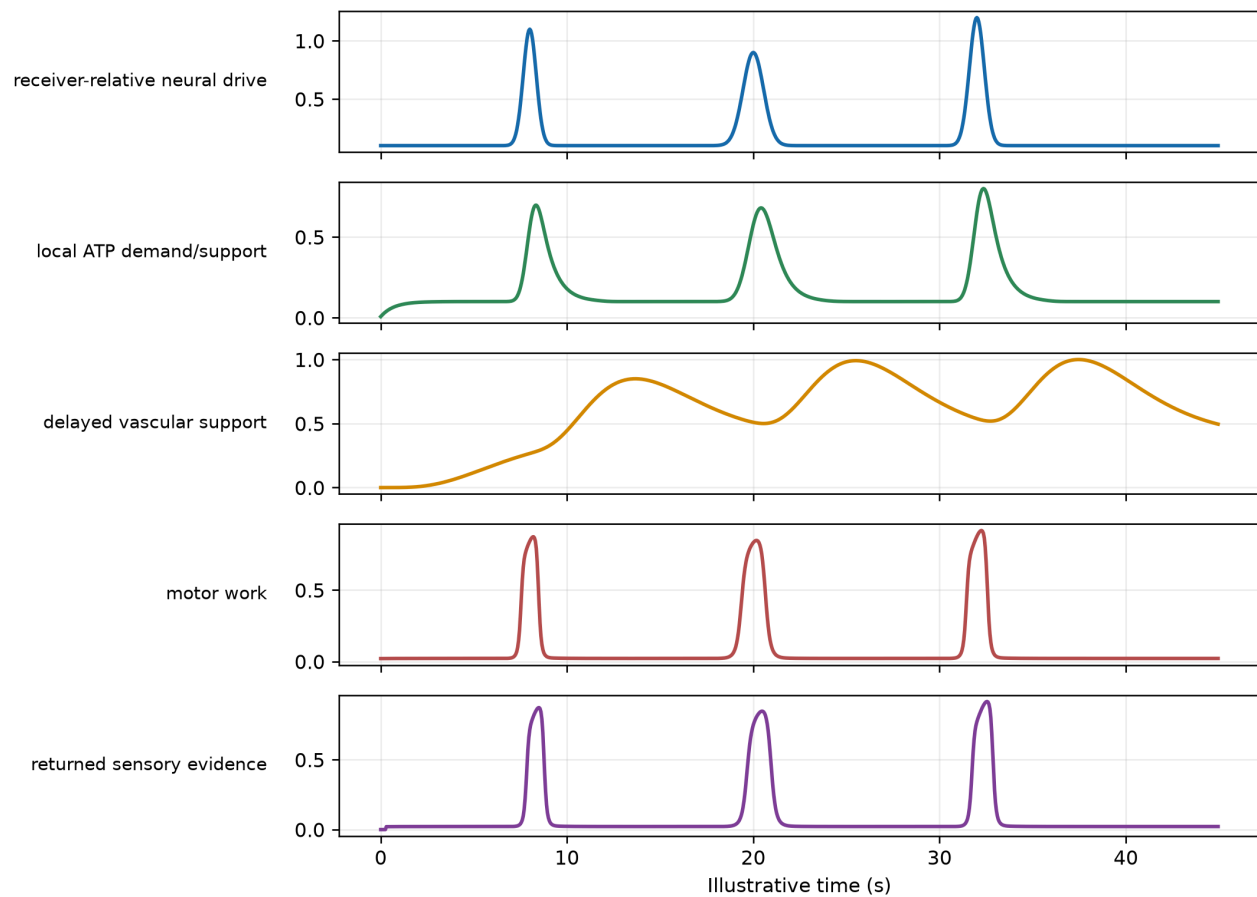


Figure 3. Illustrative fast-to-slow-to-return sequence. The traces explain delay structure only; amplitudes and time constants are not fitted to a biological dataset.

All nine prespecified model checks pass. The script, machine-readable results, metrics table, and figures accompany the manuscript. Their valid conclusion is identifiability under declared generative assumptions. Biological effect sizes require the experiments in Section 11.

13. Predictions and decision rules

The model makes seven separable predictions:

1. Receiver-relative neural valence will predict local energetic demand better than an otherwise matched model using only mean rate or broadband power.
2. Phase-aligned and phase-dispersed population patterns matched for per-cell firing will produce different ATP responses only when their receiver geometry or field relation differs in the predicted direction.
3. The established calcium and metabolic variables will explain a large share of activity-dependent ATP production; any direct field contribution will appear as a smaller incremental term.
4. Motor-unit discharge and recruitment will predict ATP cost and force through state-dependent, muscle-specific transfer functions rather than one universal proportionality.
5. Neurovascular responses will follow local demand through a delayed and pathway-dependent kernel, not at action-potential speed.

6. Perturbing one transfer interface will produce a feature-specific downstream loss, while a matched rescue restores the predicted consequence.
7. A field feature will not cross the frozen useful-effect threshold in a field-null preparation; any positive result must replicate across held-out biological units and survive selective perturbation and rescue.

These rules localize revision. A negligible field coefficient removes the direct electromagnetic route while retaining the ionic, mitochondrial, vascular, and muscular chain. Failure of the neural valence estimator narrows that receiver model without erasing activity-dependent ATP production. Failure of one vascular kernel changes the hemodynamic mapping without changing muscle energetics. The architecture therefore improves by identifying which interface succeeds and which must be replaced.

14. Consequences for PWD, neural rendering, and action

PWD becomes more biologically complete when its downstream cost is visible. A phasic departure is not merely a pattern available to a decoder. It changes ionic work, release, metabolic demand, receiver readiness, and potentially the vascular resources recruited to maintain the new state. Likewise, a tonic canvas is not metabolically inert or synonymous with one slow frequency band. It is an actively maintained context whose persistence requires cellular work.

This bridge also clarifies neural rendering. A distributed rendering has biological consequence when it changes which receivers are driven, which actions are prepared, how much cellular and muscular work is recruited, and what evidence returns. Therefore the network's internal receivers and transmitters are inseparable from the metabolic cells and bodily actuators that keep the loop running.

The same logic applies to emotion. A felt change can coincide with measurable electrophysiological, autonomic, vascular, and metabolic shifts because the organism is moving among action-relevant states. Psychological valence, SAN electrical valence, ATP-powered metabolic valence, and muscular force remain distinct variables, but transduced valence provides a disciplined way to test how they become coupled in one embodied episode.

15. Relationship to companion SAN papers

This paper forms one part of a coordinated Self Aware Networks research program. **Coherence-Weighted Electrical Drive in Choice and Motor Realization** develops the receiver-relative electrical-valence estimator and its choice-and-action perturbation protocol. **Phase-Wave Differential Calculus** formalizes the typed PWD record, Neural Tuning, Taylor and Volterra comparisons, simulations, and adverse results. **Action Potential Magnitude Effect** develops the conductance-to-waveform-to-calcium-to-release mechanism and its experimental tests. The **Formal SAN Systems Whitepaper** situates these mechanisms within the wider architecture.

The present paper supplies the cross-scale bridge: it distinguishes neural electrical drive from ATP-supported work, introduces the ATP free-energy formulation and neural-to-ATP-to-force measurement ladder, incorporates neurovascular delay, and makes the optional endogenous-field mitochondrial term independently removable and testable. Together, the papers connect local signal recognition, receiver-relative change, energetic support, motor realization, and sensory return without collapsing their distinct measured variables.

16. Scope, limitations, and failure conditions

This paper is a theoretical and computational preprint. It does not report a new biological experiment, treat its constructed model as data from a nervous system, or assign one universal conversion factor to voltage, ATP, blood flow, force, and experience. Psychological valence, receiver-relative electrical drive, biochemical power, and

action-relative force remain distinct variables whose coupling must be measured.

The neural phasor in Equation (1) is an operational feature, not automatically a literal field. The direct mitochondrial field term has a lower evidentiary status than the ionic and metabolic route. Ephaptic neuronal sensitivity and applied magnetic-field effects in NARP cybrids do not fill the missing endogenous neuronal experiment. The bat findings establish prediction and directionality patterns in their preparations but remain correlational with respect to vocal initiation.

The proposal fails locally if receiver-relative features do not improve held-out prediction, if an added field term fails the null/positive-control and perturbation/rescue sequence, if typed cross- interface models do not outperform capacity-matched disconnected models, or if the claimed delays and downstream effects do not appear in the declared measurements. Such failures revise the affected interface rather than erasing independently established ATP, vascular, or muscle biology.

17. Conclusion

A neural event becomes biologically meaningful through what it changes. It alters receiving cells, creates energetic work, recruits metabolic and vascular support, changes motor-pool activity, produces ATP-powered force, and returns a new sensory state. Neuroscience already documents many of these transformations separately. Transduced valence turns their connection into one explicit, testable program.

The central claim is that an action-relative direction and effective magnitude can be followed as living tissue transforms neural electrical drive into cellular support, vascular recruitment, and mechanical work. Neural currents, timing, routing, sign, and receiver state produce the electrical valence. ATP sustains the electrochemical machinery and powers downstream cellular and muscular work. The direct endogenous-field-to-mitochondrion proposal adds a separately estimable coupling whose contribution can be measured within the same cross-scale program.

Code and data availability

The deterministic construction, machine-readable metrics, result manifest, and three generated figures accompany this preprint. The construction uses only synthetic data and a fixed random seed. No human or animal dataset is redistributed.

Ethics statement

No new human or animal experiment was performed for this theoretical and computational paper. Future work must obtain the approvals appropriate to neural recording, stimulation, metabolic imaging, and muscular measurement in the chosen preparation.

Author contributions

Micah Blumberg conceived the framework, historical source program, terminology, and proposed experiments and wrote the manuscript.

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References

1. Basser PJ, Mattiello J, LeBihan D. MR diffusion tensor spectroscopy and imaging. *Biophysical Journal*. 1994;66(1):259-267. DOI
2. Stuart GJ, Sakmann B. Active propagation of somatic action potentials into neocortical pyramidal cell dendrites. *Nature*. 1994;367:69-72. DOI
3. Rangaraju V, Calloway N, Ryan TA. Activity-driven local ATP synthesis is required for synaptic function. *Cell*. 2014;156(4):825-835. DOI
4. Ashrafi G, de Juan-Sanz J, Farrell RJ, Ryan TA. Molecular Tuning of the Axonal Mitochondrial Ca²⁺ Uniporter Ensures Metabolic Flexibility of Neurotransmission. *Neuron*. 2020;105:678-687.e5. DOI
5. Yang W, et al. Rheb mediates neuronal-activity-induced mitochondrial energetics through mTORC1-independent PDH activation. *Developmental Cell*. 2021;56:811-825.e6. DOI
6. Anastassiou CA, Perin R, Markram H, Koch C. Ephaptic coupling of cortical neurons. *Nature Neuroscience*. 2011;14:217-223. DOI
7. Hall DN, et al. Capillary pericytes regulate cerebral blood flow in health and disease. *Nature*. 2014;508:55-60. DOI
8. Mishra A, et al. Astrocytes mediate neurovascular signaling to capillary pericytes but not to arterioles. *Nature Neuroscience*. 2016;19:1619-1627. DOI
9. Isaacs D, Xiang L, Hariharan A, Longden TA. KATP channel-dependent electrical signaling links capillary pericytes to arterioles during neurovascular coupling. *Proceedings of the National Academy of Sciences*. 2024;121(50):e2405965121. DOI
10. Weineck K, Garcia-Rosales F, Hechavarria JC. Neural oscillations in the fronto-striatal network predict vocal output in bats. *PLOS Biology*. 2020;18(3):e3000658. DOI
11. Garcia-Rosales F, et al. Echolocation-related reversal of information flow in a cortical vocalization network. *Nature Communications*. 2022;13:3642. DOI
12. Christie AD, Foulis SA, Kent JA. ATP cost of muscle contraction is associated with motor unit discharge rate in humans. *Neuroscience Letters*. 2016;629:186-188. DOI
13. Del Vecchio A, et al. You are as fast as your motor neurons: speed of recruitment and maximal discharge of motor neurons determine the maximal rate of force development in humans. *The Journal of Physiology*. 2019. DOI
14. Ito M, et al. Extremely low-frequency electromagnetic field (ELF-EMF) enhances mitochondrial energy production in NARP cybrids. *Scientific Reports*. 2025;15:24369. DOI
15. Blumberg M. Internally dated 2017 tonic/phasic, electric-field, and echolocation source; public Git form fixed in 2022. *Self-Aware Networks*. Git source
16. Blumberg M. Action Potential Magnitude Effect development source. *Self-Aware Networks*. 2022. Git source
17. Blumberg M. Phase Shifts as Electromagnetic Wave Valence. *Self-Aware Networks*. 2022. Git source
18. Blumberg M. Bat information-flow integration. *Self-Aware Networks*. 2022. Git source
19. Blumberg M. NAPOT/PWD synthesis and bat top-down-control interpretation. *Self-Aware Networks*. 2022. Git source
20. de Polavieja GG, Harsch A, Kleppe I, Robinson HPC, Juusola M. Stimulus history reliably shapes action potential waveforms of cortical neurons. *The Journal of Neuroscience*. 2005;25(23):5657-5665. DOI
21. Blumberg M. Learning-loop dialogue preserved in a0072z; authored as occurring before September 21, 2011 and publicly Git-fixed in 2022. *Self-Aware Networks source archive*. Git source
22. Blumberg M, Andrews M. Neo Mind Cycle and My Happiness Makeover. Public dialogue uploaded August 1, 2012 local time. Public video
23. Blumberg M. The Neural Lace Podcast: original monologue. Public audio uploaded April 11, 2017. Public audio